

Physical Evidence



Learning Objectives

- List the common types of physical evidence encountered at crime scenes.
- Explain the difference between the identification and comparison of physical evidence.
- Define individual and class characteristics and give examples of physical evidence having these characteristics.
- List the common databases and describe how they are used.

Physical Evidence: any object that can establish that a crime has been committed or can link a crime and its victim to perpetrator



Evidence is presented at trial and may consist of:

- Testimony of witnesses
- Introduction of records or documents
- Introduction of exhibits or objects (physical evidence)

Evidence is offered for the purpose of assisting the jury or judge (Trier-of-fact)



Circumstantial evidence – based on inference and not personal knowledge or observation

Corroborating evidence – evidence that differs but strengthens or confirms other evidence

Exculpatory evidence – tends to establish innocence of defendant

Probative evidence – tends to prove or disprove a point

Example: If a man accused of embezzling money from his company had made several big-ticket purchases in cash around the time of the alleged embezzlement, that would be _____ evidence that he had stolen the money.



Common Types of Physical Evidence:

Biological: blood, semen, saliva, plant material, wood, bones

Chemical: accelerants, unknown substances, glass, GSR, drugs (legal and illicit), paper, inks, metals, polymers, paint, tape toxicology specimens

Trace: hairs, fibers, soil

Impression: fingerprints, firearms, handwriting, indented writing, serial number restoration, footwear, tire treads, tool marks

Digital: cell phones, computers, cameras, pagers



Biological Evidence -

Forensic Serology

Forensic Molecular Biology (DNA)

What is the nature of the biological fluid? Blood, semen, saliva, urine

Is it human or from another species?

Does the evidence sample have sufficient DNA to compare to a known sample?

Is the DNA profile from the evidence sample identical to the suspect or victim?



Chemical Evidence –

Forensic Chemistry

Forensic Toxicology

What is the substance?

Is there a foreign substance present?

How much is present?

What is the quality of the substance?

Is it similar to known standards?
(tapes, paint, ink)



Trace Evidence –

Forensic Microscopy

Forensic Geology/Mineralogy

Is this a hair? Is it human? What part of the body did it come from?

Can the hair be included or excluded as coming from the suspect?

Is the soil in the tire tread similar to soil from the crime scene?

What are the properties of the recovered fiber? Are they consistent with the properties of the fiber from the suspect's clothing?



Trace evidence is microscopic evidence

Impression Evidence –

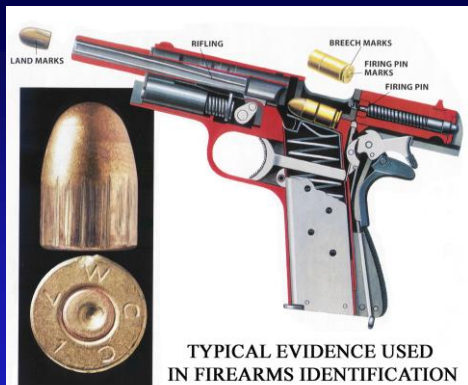
Firearms and Toolmarks
Fingerprints
Handwriting
Shoeprints
Tire treads



What type of weapon fired the bullet recovered from the victim? Did this weapon fire this bullet? What was the distance from the weapon to the victim?

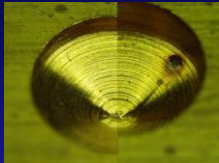
Did the fingerprint recovered from the crime scene have the same pattern and features as one of the suspect's fingerprints?

Did the suspect sign this document?



Toolmarks -

Was this drill bit used to open the lock?



Can an obliterated serial number be restored?



Fingerprints –

- Are fingerprints present?
- Are they from the suspect?
- Can we compare the prints to a database?



Impression Evidence

- Impression evidence is usually found at crime scenes in the form of shoe and tire marks.
- They may be three -dimensional when made in soft dirt, or two-dimensional when left on a hard surface.
- Other forms of impression evidence: toolmarks and bite marks.

Shoe Prints Tire Marks Bite Marks



Digital Evidence - a process of searching and Analyzing digital devices for evidence of a crime.



Class Characteristics

Class characteristics – are not unique to an object but allow an item to be included or excluded as belonging to another group.

Characteristics include size, shape, color, manufacturing patterns, etc.



Class Characteristics

Example: Carpet fiber

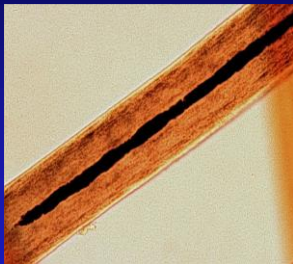
Does the fiber taken from the victim's body have properties consistent with the carpet found in the suspect's home? (Can it be included or excluded?)



Comparison – the questioned evidence is compared with objects whose source is known resulting in an inclusion or exclusion result; utilizes class characteristics

Hairs
Fibers
Soil
Unknown powder
Unknown liquid
Automotive paint

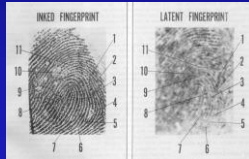
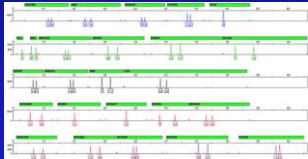
If objects do not share similar characteristics, then they are of a different origin = exclusion



Individual Characteristics

Individual characteristics – unique characteristics that allow an item to be attributed to a specific item or person; exist in both the known and questioned samples.

Example: Fingerprints and DNA can identify the specific individual who deposited the print or body fluid.



Identification - the determination of the identity of a substance, object, or person with as near absolute scientific certainty as technique will permit; requires individual characteristics

Fingerprints
Nuclear DNA
Firearms and toolmark striations
Irregular footwear
Broken, torn or cut edges



Class or Individual Characteristic?

Caucasian human head hair
Red cotton fiber
9mm bullet with a right twist
Microscopic striations on bullet
Whorl fingerprint pattern
Signature
Broken piece of glass -
chemical and physical
fracture match

DATABASES

PDQ – automotive paint

CODIS – DNA, missing persons

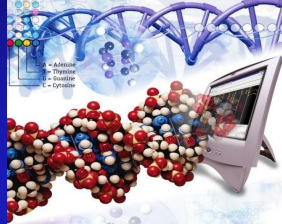
NGI – fingerprints

NIBIN – firearms

Carpet fiber

Footwear

EXPeRT



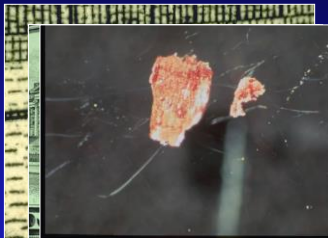
Paint Analyses

- Types of paint examined include architectural, tool, automotive and other vehicles.
- The most information is usually obtained from automotive paints due to unique combinations of specialty layers and additives.



Paint Analyses

Through the use of microscopy and highly sensitive instrumentation, even the smallest paint chips may be analyzed.



Paint Analysis

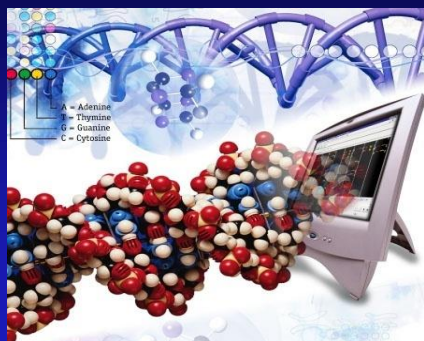
Using the FBI Laboratory's National Automotive Paint File (NAPF) and the Royal Canadian Mounted Police Laboratory's Paint Data Query (PDQ), the examiner can often determine the make, model, and year of an automobile from a single paint chip left at a crime scene.



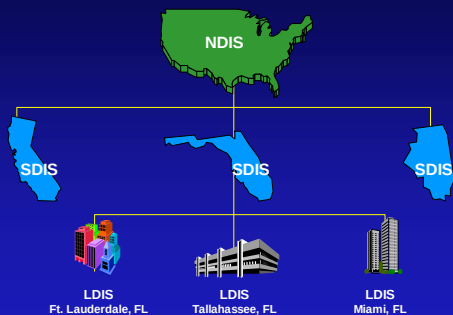
NAPF – USA vehicles

PDQ – USA + World vehicles

CODIS: Combined DNA Index System

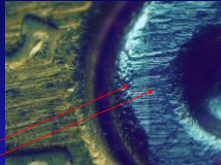


FBI's CODIS Architecture



NIBIN – National Integrated Ballistics Information Network

Identification of a breech face



Latent Print Database

FBI's IAFIS (Integrated Automated Fingerprint Identification System)

Over 50 million individual's fingerprints in criminal database. All fingerprints and most criminal history records automated.

This system was replaced by NGI in September 2014

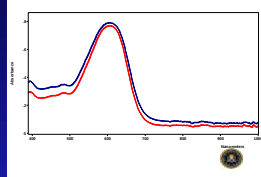
Next Generation Identification System

NGI Capabilities:

- Advanced Fingerprint Identification Technology (AFIT) *
- Latents and National Palm Print System (NPPS)
- Rap Back
- Repository for Individuals of Special Concern (RISC)
- Interstate Photo System
- Iris Recognition (IR)

* Improved matching accuracy from 92 % to more than 99.6 %; faster response times, fewer transaction rejects, and increased frequency of identification

Automotive Carpet Fiber Database



Two instruments used to determine class characteristics.

- Microspectrophotometer – fiber color
- Fourier Transform Infrared Spectroscopy (FTIR) – fiber chemical composition

Footwear Database



Consists of collection of sole and heel designs

Enables determination of manufacturer of shoes from prints or impressions left at crime scene



Example of image in the file

FBI's Explosives Reference Tool Database



Searchable document repository

Forensic tool used to assist law enforcement in Terrorism and Bombing investigations

EXPeRT Database

Product information (Open Source)

Technical and formulation information (Proprietary)

High quality photographs of products



Database Search “TATP”

